

Why Quantum Mechanics in Biophysics?

Introduction to the Quantum World

PWF Summer school on Theoretical and Computational Biophysics

University of Colombo

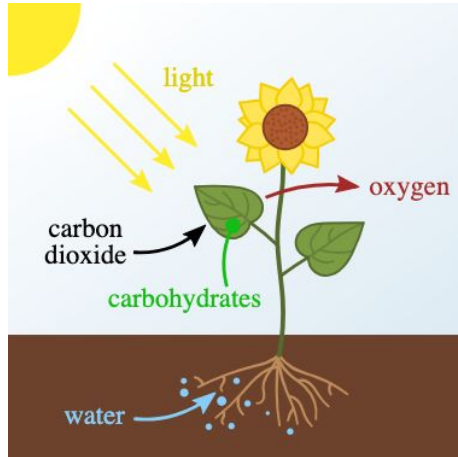
Day 4/5

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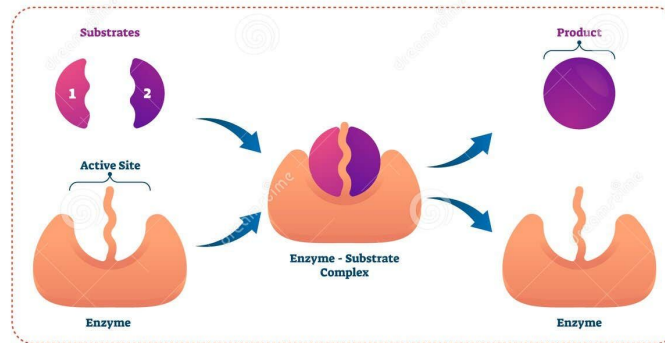
Why quantum mechanics in biophysics?

Several key processes in biology and chemistry depend on the **quantum-mechanical behavior of electrons** — bond making and breaking, charge redistribution, and interactions that classical force fields cannot capture.

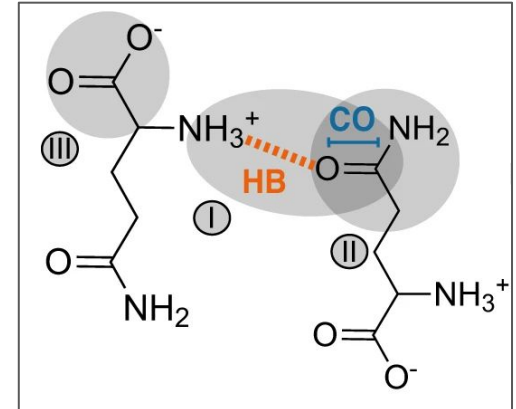
Photosynthesis



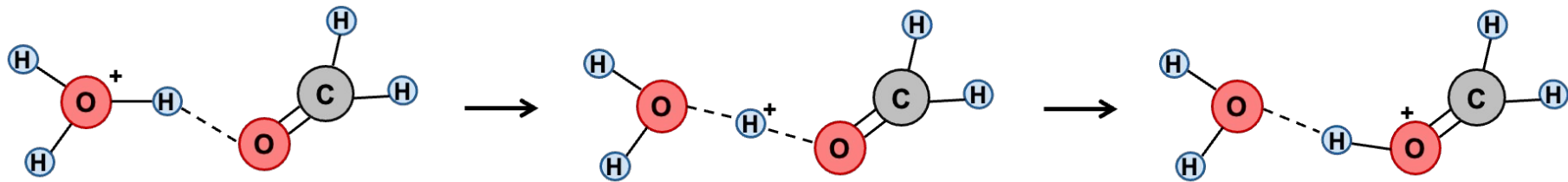
Enzyme catalysis



Proton transfer mechanisms



Limitations of classical force fields



Can we follow this reaction with classical MD?

- Standard FFs describe atoms as classical particles connected by springs → no electrons in the model
- The bonding topology is usually fixed at input and cannot change during the simulation → bond breaking/forming is inaccessible by construction
- Electronic response is only approximated → no explicit description of polarization or charge transfer

First-principles description of the electronic structure

The electronic-structure problem starts from the time-independent **Schrödinger equation**:

Hamiltonian operator:
it represents the total energy operator of the system

$$\hat{H}\Psi = E\Psi$$

Wavefunction:
it describes the quantum state of the system

Eigenvalue of $\hat{H}$:
the total energy of the state Ψ

The diagram shows the Schrödinger equation $\hat{H}\Psi = E\Psi$. A horizontal arrow points from the text 'Hamiltonian operator: it represents the total energy operator of the system' to the operator $\hat{H}$. A vertical arrow points from Ψ down to the text 'Wavefunction: it describes the quantum state of the system'. A diagonal arrow points from E down and to the right to the text 'Eigenvalue of $\hat{H}$: the total energy of the state Ψ '.

- If Ψ is an **eigenfunction** of $\hat{H} \rightarrow$ when $\hat{H}$ acts on Ψ , it returns the same Ψ , just multiplied by the number E (= **eigenvalue**).
- The lowest-energy eigenstate Ψ_0 with energy E_0 is called the **ground-state** of the system

A few properties of quantum states

Properties of the wavefunction

$$\langle \Psi_i | \Psi_j \rangle = \delta_{ij}$$

Kronecker delta

- Each state is **normalized**
- Different eigenstates are **orthogonal**

Variational principle

$$E[\Psi_{trial}] = \frac{\langle \Psi_{trial} | \hat{H} | \Psi_{trial} \rangle}{\langle \Psi_{trial} | \Psi_{trial} \rangle} \geq E_0$$

- Any trial wavefunction (Ψ_{trial}) gives an energy equal to or higher than the true **ground-state** energy (E_0)
- The exact ground-state wavefunction (Ψ_0) **minimizes** the energy

The molecular problem

For a molecular system of N_{el} electrons and N_{nucl} nuclei, the Hamiltonian splits into kinetic and potential contributions:

$$\hat{H} = \hat{T}_N + \hat{T}_e + \hat{V}_{NN} + \hat{V}_{ee} + \hat{V}_{eN}$$

$$\hat{H} = - \sum_I \frac{\hbar^2}{2M_I} \nabla_I^2 - \sum_i \frac{\hbar^2}{2m_e} \nabla_i^2 + \sum_{I < J} \frac{Z_I Z_J e^2}{4\pi\epsilon_0 R_{IJ}} + \sum_{i < j} \frac{e^2}{4\pi\epsilon_0 r_{ij}} - \sum_{iI} \frac{Z_I e^2}{4\pi\epsilon_0 r_{iI}}$$

Nuclear kinetic energy	Electronic kinetic energy	Nuclear-nuclear repulsion	Electron-electron repulsion	Nuclear-electron attraction
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- Nuclei and electrons are coupled through $V_{eN} \rightarrow$ the many-body Ψ can't be separated
- Exact solution is out of reach for any real molecular system

Born-Oppenheimer approximation

Can we approximately decouple nuclei and electrons?

- Nuclei are much heavier than electrons: $m_{\text{nucl}} \gg m_{\text{el}}$
- At a given kinetic energy, electrons move much faster than nuclei $\rightarrow$ on the timescale of nuclear motion, electrons “instantaneously” adjust to the current nuclear positions



We can fix the nuclei at positions $\mathbf{R}$ ($T_N = 0$) and solve the electronic Schrödinger equation:

$$\hat{H}_{el} \Psi_{el}(\mathbf{r}; \mathbf{R}) = E_{el}(\mathbf{R}) \Psi_{el}(\mathbf{r}; \mathbf{R})$$

$$\hat{H}_{el} = \hat{T}_e + \hat{V}_{ee} + \hat{V}_{eN}(\mathbf{r}; \mathbf{R})$$

Born-Oppenheimer approximation

$$\hat{H}_{el}\Psi_{el}(\mathbf{r};\mathbf{R}) = E_{el}(\mathbf{R})\Psi_{el}(\mathbf{r};\mathbf{R})$$



R is now a parameter: for every nuclear geometry, we solve a different electronic problem.

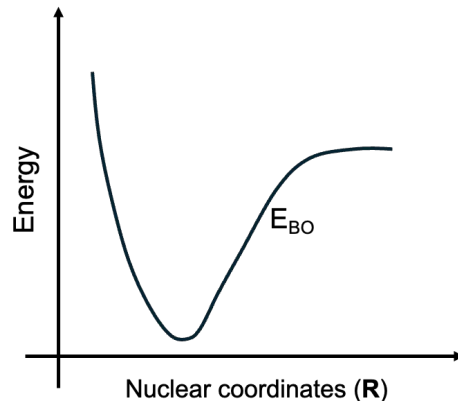
If we then add the nuclear repulsion:

$$E_{BO}(\mathbf{R}) = E_{el}(\mathbf{R}) + V_{NN}(\mathbf{R})$$

E_{BO} = Potential-energy surface on which the nuclei move



We'll come back to this in lecture 12!



Solving the electronic problem

We still need to solve the many-electron electronic problem → exact solution is out of reach but we can use approximated methods:



Wavefunction-based methods

- Central object: many-electron Ψ
- Simplest approximation: **Hartree-Fock** (HF)
- In HF, Ψ is approximated by a single Slater determinant; each electron moves in the mean field of the others
- Missing: electron correlation beyond exchange
- Post-HF methods improve accuracy but scale steeply with system size ($\propto N^5$ to N^7)



Density-based methods

- Central object: electron density ρ
- **Density functional theory** (DFT)
- Exchange and correlation are described through the exchange-correlation functional
- Typically more favorable scaling ($\sim N^3$) → applicable to large systems
- Widely adopted in biophysics/computational chemistry